

# Generation Interconnection System Impact Study Report

## For Queue Project Q18-1058

Joppatowne 138 kV

March 04, 2020

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1058. The Interconnection Customer proposes a 165.6 MW solar photovoltaic facility interconnecting at the Joppatowne 138 kV substation in the Maryland area. The request was received on September 07, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 358 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$556,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: BARLING / JOPPATOWNE 138.0 kV  
Contingency: JOPPATOWNE / BARLING 138.0 kV Cir. 2  
Loading: 111.6% (Rating: 464 MVA)  
DFAX: 65.4%  
Pre-project loading: 88.3%
2. Facility: JOPPATOWNE / BARLING 138.0 kV Cir. 2  
Contingency: BARLING / JOPPATOWNE 138.0 kV Ckt 1  
Loading: 111.6% (Rating: 464 MVA)  
DFAX: 65.4%  
Pre-project loading: 88.3%

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                             | Required Upgrade                            | Cost Estimate | Allocation to Project |
|--------------------------------------|---------------------------------------------|---------------|-----------------------|
| BARLING / JOPPATOWNE 138.0 kV        | Rebuild line with higher-capacity conductor | \$1,984,000   | 14% (\$271,000)       |
| JOPPATOWNE / BARLING 138.0 kV Cir. 2 | Rebuild line with higher-capacity conductor | \$2,087,000   | 14% (\$285,000)       |
|                                      | Total allocated to project                  |               | \$556,000             |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.